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Article

Keywords:

Posted Date: January 16th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-8376654/v1>

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Additional Declarations: There is **NO** Competing Interest.

The global potential of aquavoltaics for power generation and offsetting the carbon footprint of freshwater aquaculture

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Abstract

Rising pressures at the energy–food–climate nexus are driving demand for more sustainable production systems. Global inland freshwater aquaculture has rapidly expanded, intensifying greenhouse gas emissions and sustainability challenges. As a potential solution, aquavoltaics—integrating photovoltaics with aquaculture—offers multifunctional benefits, but its global deployment and decarbonization potential remain poorly understood. Here, we integrate satellite observation products with realistic climate-driven photovoltaic system simulations to map the global expansion patterns and installation potential of aquavoltaics. The results revealed a current global aquavoltaics area of 423 km² by 2023, with China accounting for the overwhelming majority (97%). We estimate that deploying PV with a 40% coverage rate across over 50,000 km² of landside clustering of aquaculture ponds globally could yield an electricity generation potential of $4,143 \pm 145$ TWh yr⁻¹, sufficient to support energy self-sufficiency in many cities and offset freshwater aquaculture carbon footprint in numerous countries. These findings highlight aquavoltaics as a scalable pathway towards sustainable aquaculture and low-carbon energy futures.

Main

A central challenge in the era of energy–food–climate transitions is to secure both low-carbon energy and sustainable food production under growing population and climate pressures. Driven by rising demand for animal protein, global consumption of aquatic foods (excluding algae) has increased substantially, growing more than five-fold over the past six decades to around 162 million tonnes (Mt) in 2021 (by live weight)^{1,2}. With wild capture fisheries plateauing in recent decades, aquaculture has become essential for meeting future food demands and strengthening the resilience of global food systems^{3,4}. However, driven by the intensification and spatial expansion, freshwater aquaculture development has become a significant contributor to greenhouse gas (GHG) emissions⁵⁻⁷. The carbon footprint of freshwater aquaculture arises mainly from feed production, energy use, and emissions from aquatic environments, contributing an estimated 377 ± 100 Tg CO₂-equivalent (CO₂e) in 2012⁸. Reliance on freshwater systems to feed a growing population could lead to a 54–64% increase in GHG emissions by 2050 under various socio-economic scenarios⁸, intensifying global climate challenges. Consequently, the sector faces mounting urgency to adopt comprehensive sustainability measures to reconcile its critical role in food security with the imperative to mitigate its environmental impact.

Aquavoltaics—the co-location of photovoltaic (PV) systems over aquaculture ponds—offers a promising solution to the pressing global challenges of energy–food–climate transitions by addressing both sustainable food production and low-carbon energy generation. By integrating renewable energy production into aquaculture systems, aquavoltaics minimizes land use competition, making it particularly suitable for regions where land scarcity and high costs constrain the deployment of terrestrial PV systems⁹⁻¹¹. Given the extensive global coverage of land-based aquaculture ponds, aquavoltaics presents significant potential for PV power generation. One such analysis estimated that if the existing 2,000 km² of freshwater aquaculture area in the United States were utilized for aquavoltaics, the resulting power generation could account for approximately 10.3% of the country’s total energy consumption in 2016¹¹. Studies also indicate that with a well-designed PV coverage ratio, shading effects have minimal negative impact on fish yields, with the significant economic returns from energy generation offsetting any

marginal losses¹²⁻¹⁴. Additionally, aquavoltaics offer multiple co-benefits such as reduced water evaporation, lower PV panel temperatures, and improved energy conversion efficiency^{11,15}. These advantages have spurred global promotion of aquavoltaics, particularly in Asia, where vast aquaculture areas have enabled rapid development, led by China^{10,16}.

Despite the rapid expansion of aquavoltaics, its global development status and potential, particularly for power generation and carbon footprint mitigation, remain poorly understood. Compared to FPV systems on reservoirs and lakes, which rely on floating platforms to accommodate dynamic water levels, aquavoltaics are typically deployed in shallow, stable water bodies. It prioritizes flexible configurations of fixed-tilt structures with variable density and tilt angles to optimize cost efficiency^{10,17}. Traditional approaches for assessing water-surface PV (WSPV) generation potential, which adopt configurations characteristic of low-tilt-angle and high-density FPV installations, are therefore not well suited for these systems¹⁸⁻²¹. Additionally, in countries like China, stricter regulatory oversight has been applied to the approval of WSPV projects other than aquavoltaics²², as large-scale PV installations on open water bodies are often perceived to compromise hydrological integrity and flood management functions. However, the regulatory requirements for aquavoltaics may be less stringent than those for other types of WSPV on open water bodies. As the world's largest aquaculture producer, the Chinese government has been actively encouraging aquavoltaic projects by offering subsidies or other incentives, provided that these projects do not compromise the safety of aquaculture production. In light of concerns regarding the impact of other types of WSPV on open water ecosystems, as well as the growing interest in aquavoltaics, a systematic evaluation of aquavoltaics' global potential for energy production and its role in offsetting the carbon footprint of freshwater aquaculture is crucial for advancing sustainable energy and food systems.

In this study, we integrated recent global WSPV datasets^{23,24} with multi-source geospatial data to map the spatial distribution and temporal changes of aquavoltaics across the globe, characterizing its current development landscape. Drawing on advancements in high-resolution global aquaculture ponds mapping²⁵, we analyzed over 50,000 km² of landside clustering of aquaculture ponds (LCAP), encompassing more than 250,000 spatial polygons. We then quantified their power generation potential under realistic climate conditions and optimal fixed-tilt configurations, with the PV system performance model (PVLIB-Python, Sandia National Laboratories)²⁶, assuming a conservative PV coverage ratio of 40%. Building on this, we further estimated the minimal proportion of aquaculture ponds needed to fully offset the carbon footprint of freshwater aquaculture across major aquaculture-producing countries. This analysis provides insights into sustainable aquaculture development pathways by guiding low-carbon strategies and facilitating the sector's green transition. All computational methods and assumptions are detailed in the Methods section.

Results

Spatiotemporal distribution of aquavoltaics

Our analysis indicates that, as of 2023, aquavoltaics spanned approximately 423 km² globally, comprising 73% of the total area of WSPV. The aquavoltaics were mainly deployed in 15 countries—11 of which are in Asia—with a heavy concentration in China, accounting for 97% of the global aquavoltaics area

(Supplementary Table 1). In several coastal provinces of China, such as Anhui and Guangdong, aquaculture ponds have become the dominant land-use type for newly deployed PV plants (Supplementary Fig. 1). Other notable contributors include Vietnam (0.9%), South Korea (0.7%), and Thailand (0.4%) (Fig. 1a, c, d).

The earliest detectable aquavoltaics dates back to 2010. From 2012 to 2016, aquavoltaics experienced a phase of steady growth, with an average annual expansion of 16 km² ($R^2 = 0.98$). A pronounced acceleration followed in 2017, with deployment rates tripling to 48 km² per year ($R^2 = 0.96$). Throughout this period, aquavoltaics consistently expanded at a faster rate than other forms of WSPV deployed in open water bodies such as lakes and reservoirs, reflecting their substantial potential for renewable energy development and efficient use of water-based resources. In 2023, the newly added aquavoltaics area was nearly six times larger than that of other WSPV types (Fig. 1b). While China is driving the global expansion of aquavoltaics, other regions have also expanded rapidly, with their total area increasing nearly fivefold since 2018.

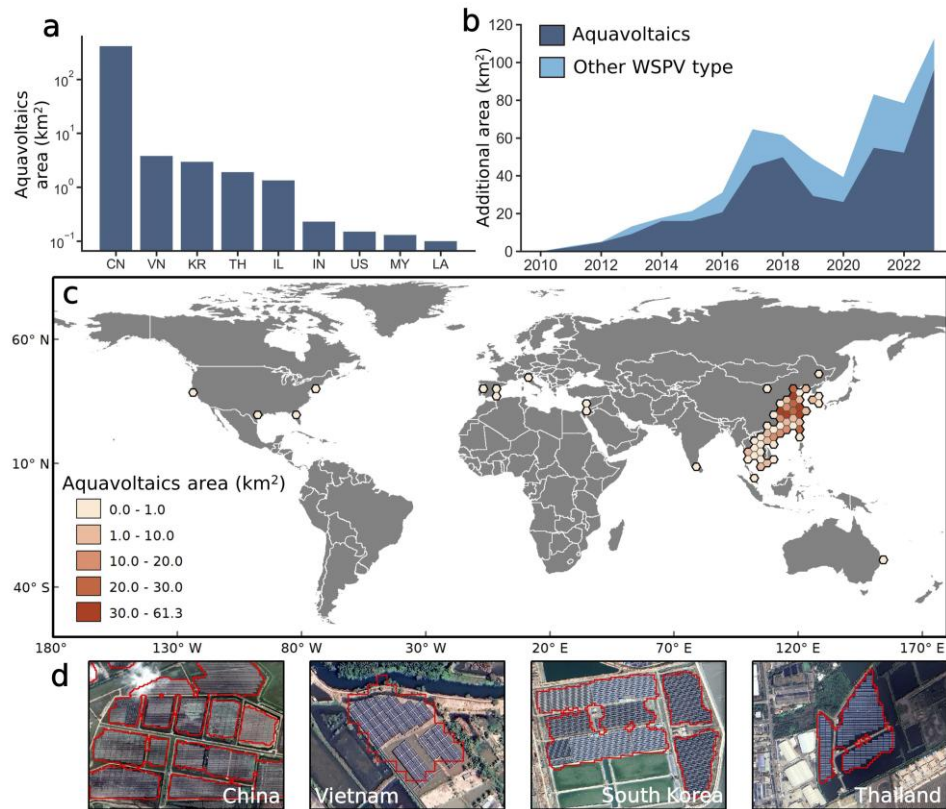


Fig. 1 Spatiotemporal distribution of aquavoltaics. (a) Aquavoltaics area in major countries (only countries with aquavoltaics area >0.02 km² are shown; country names are abbreviated using ISO 3166-1 two-letter country codes). (b) Annual newly installed area of aquavoltaics compared to other WSPV types installed on non-aquaculture water bodies (e.g., lakes and reservoirs). (c) Spatial distribution of global aquavoltaics across a 3-degree hexagonal grid. (d) Aquavoltaics examples in the top four countries by area (red lines delineate aquavoltaics boundaries within the dataset).

Global power generation potential for aquavoltaics

Since the mid-1970s, there has been a pronounced expansion, intensification and spatial clustering of aquaculture in coastal regions around the world ^{25,27}, increasing its spatial footprint and creating new opportunities for aquavoltaic deployment. To evaluate this potential, we adopted a 40% PV coverage ratio (defined as the ratio of PV panel projection area to aquaculture pond area), consistent with regulatory limits established in countries where explicit coverage requirements for agrivoltaic and aquavoltaic systems are in place ^{13,28}. Such limits are commonly implemented to avoid substantial reductions in agricultural or aquaculture yields. Under this assumption, we estimate that the mean power generation potential of global LCAP is $4,143 \pm 145$ TWh yr⁻¹ (mean $\pm$ standard deviation; the standard deviation primarily reflects interannual variability from 2001 to 2023 in solar energy). To examine the sensitivity of power generation potential to deployment intensity, we further evaluated a range of coverage levels, with particular focus on two alternative scenarios (Supplementary Figs. 2–4). A high-coverage scenario (60%), as proposed in previous research to maintain approximately 70% of baseline fish yields ¹⁶, yields an upper-bound potential of $5,769 \pm 199$ TWh yr⁻¹. In contrast, a low-coverage scenario (20%), consistent with ecological and operational constraints commonly applied to FPV systems ¹⁹, produces a reduced yet still considerable potential of $2,088 \pm 73$ TWh yr⁻¹.

In contrast to the high generation potential of FPV on broadly distributed reservoirs and lakes worldwide ^{18,19}, power generation potential for aquavoltaics exhibits pronounced regional disparities driven by the distribution of inland aquaculture (Fig. 2a, b, c). Power generation potential from aquavoltaics in Europe remains low because of relatively limited area of inland aquaculture (Fig. 2d and Supplementary Table 2). By contrast, power generation from aquavoltaics is predominantly concentrated in Asia, where inland aquaculture is highly developed, accounting for over 90% of the global power generation potential. China leads with the highest power generation potential ($1,321 \pm 68$ TWh yr⁻¹), aligned with its large area of LCAP (21,716 km²). Substantial contributions also arise from Southeast and South Asian countries, including Vietnam, Indonesia, India, Bangladesh, and Thailand, each surpassing 150 TWh yr⁻¹. In Africa, Egypt, the continent's leading aquaculture producer, has a power generation potential of 128 ± 1 TWh yr⁻¹. In the Americas, countries including the United States, Brazil, and Mexico each exceeds 20 TWh yr⁻¹. Nine developing countries in Asia (Vietnam, Indonesia, Bangladesh, Myanmar, Philippines, and Cambodia) and the Americas (Ecuador, Honduras, and Nicaragua) exhibit power generation potential from aquavoltaics exceeding their annual electricity demand (Supplementary Table 2). These countries often experience weak power infrastructure, limited grid coverage, and unequal access to electricity ²⁹. For both larger consumers, such as Indonesia (297 TWh yr⁻¹) and Vietnam (229 TWh yr⁻¹), and smaller economies, such as Honduras (11 TWh yr⁻¹) and Cambodia (12 TWh yr⁻¹), aquavoltaics could theoretically satisfy their entire electricity demand.

Power generation potential per unit area for aquavoltaics serves as a proxy for generation efficiency or energy yield performance, offering a complementary perspective to total generation potential. Spatial heterogeneity in power generation potential per unit area of aquavoltaics is largely attributable to regional differences in solar irradiance, which are shaped by latitude, elevation, and prevailing cloud conditions (Fig. 2d and Supplementary Fig. 5). Middle Eastern countries, including Saudi Arabia, Oman, and Yemen, exhibit high generation potential per unit area, typically exceeding 250 kWh m⁻² yr⁻¹. In contrast, Eastern Europe and East Asia, including Ukraine, Lithuania, China, and South Korea, show lower generation

potential per unit area, generally below $170 \text{ kWh m}^{-2} \text{ yr}^{-1}$. Notably, China, despite accounting for nearly 40% of global LCAP area, has a power generation potential per unit area of $155.19 \pm 8.00 \text{ kWh m}^{-2} \text{ yr}^{-1}$, only 76% of the global average.

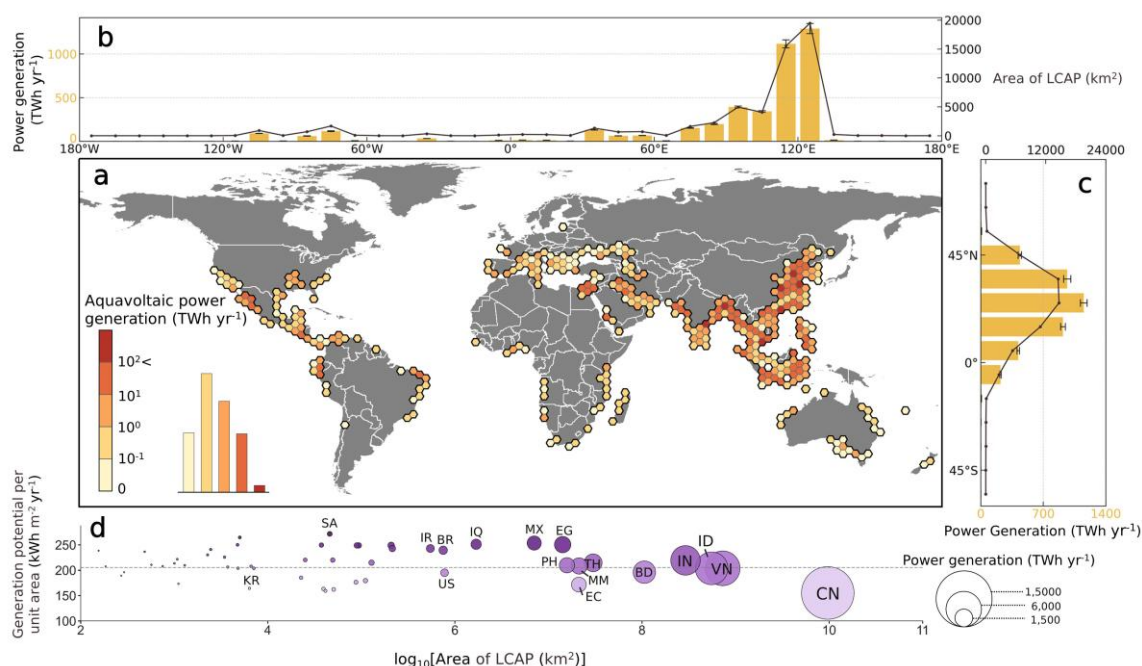


Fig. 2 Global power generation potential for aquavoltaics at a coverage of 40%. (a) Mean annual power generation potential for aquavoltaics from 2001 to 2023 across a 3-degree hexagonal grid. The inset bar chart shows the proportion of grid within each interval of potential generation from aquavoltaics. Distribution of power generation potential for aquavoltaics across (b) longitude and (c) latitude. (d) Power generation potential, area of LCAP, and power generation potential per unit area of aquavoltaics across countries. The dashed line represents the global average power generation potential per unit area. Bubble color intensity indicates the magnitude of potential per unit area. Country names are abbreviated using ISO 3166-1 two-letter country codes.

A primary benefit of aquavoltaics lies in its proximity to densely populated agricultural and fishery regions with high electricity demand. This makes aquavoltaics ideal for communities or small-to-medium cities seeking localized power generation, reducing dependence on traditional grids while promoting agricultural and rural economic development¹⁹. Our analysis shows that 878 cities across 52 countries could achieve electricity self-sufficiency with aquavoltaics and adequate storage (Fig. 3). In 2017, the total electricity consumption of these cities was 577 TWh, while the potential electricity generation from aquavoltaics is estimated to be six times higher than the apparent demand at $3476 \pm 114 \text{ TWh yr}^{-1}$. Among these cities, 56% are located in Southeast Asia, a region facing energy provisioning challenges – including high electricity prices and frequent power outages³⁰. Aquavoltaic microgrid systems could play an important role in alleviating persistent power shortages and meeting the growing energy demand driven by population growth, all without occupying additional land resources. In China, aquavoltaics could potentially meet the electricity demand of 32 cities, spanning annual electricity consumption from 1 to 39 TWh³¹. These cities range from those with low demand and economies centered on agriculture and light industry (e.g., Tianmen and Xiantao) to energy-intensive industrial hubs such as Tangshan and Dalian. Moreover, among the 152 freshwater aquaculture cities worldwide where PV plants had been

deployed by 2018, 78% sourced more than half of their PV land area from cropland ³², posing a serious threat to food security ³³. In this context, electricity generated by aquavoltaics can also limit the expansion of PV infrastructure onto adjacent cropland, thereby contributing to the protection of food security.

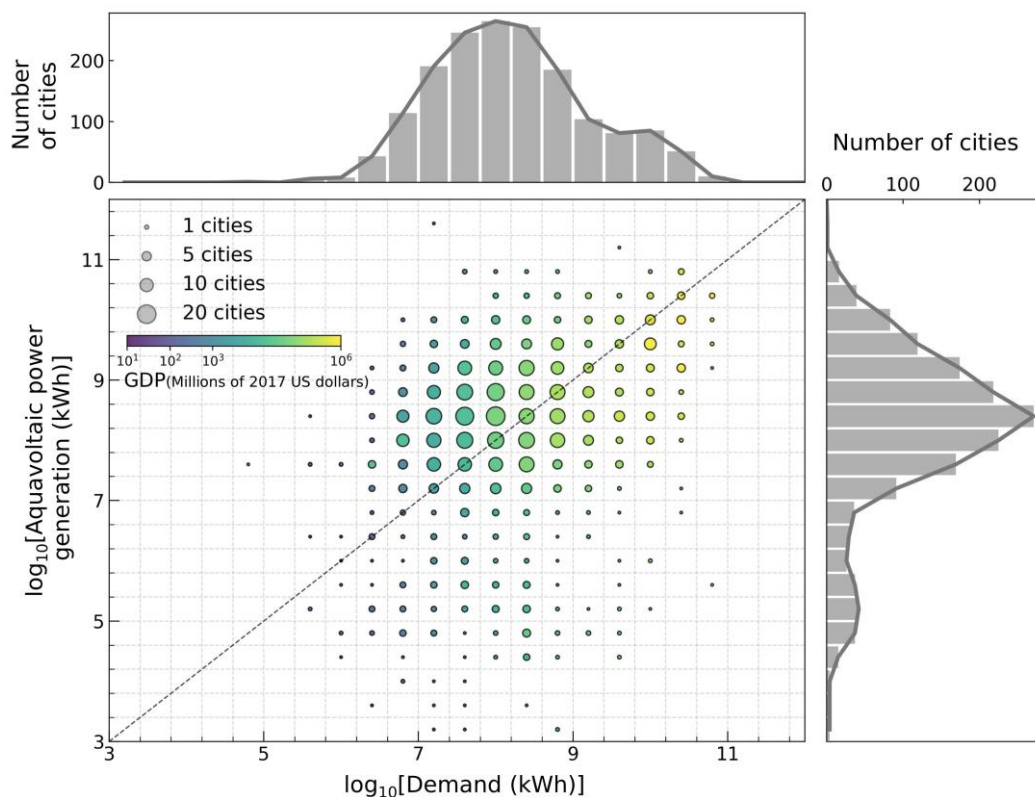


Fig. 3 Assessment of power generation potential for aquavoltaics relative to electricity demand in cities. Cities are grouped into grids at 0.4 $\log_{10}$ kWh intervals based on potential power generation versus demand. Circle size and color indicate the number of cities and the total gross domestic product of cities within each grid, respectively. The bar chart illustrates the distribution of cities across varying electricity demand and potential aquavoltaic power generation.

Potential of aquavoltaics for offsetting the carbon footprint of freshwater aquaculture

Our estimations indicate that global inland freshwater aquaculture emitted approximately 438 million tons of CO₂e to the atmosphere in 2022, encompassing GHG emissions from feed production, energy use, and the aquatic environment. This accounts for about 0.8% of total anthropogenic GHG emissions in that year ³⁴. Deploying PV systems on inland aquaculture ponds offers a viable pathway to offset the carbon footprint of freshwater aquaculture by supplying renewable electricity to the grid. Under the assumption of 40% coverage, global deployment of aquavoltaics could reduce GHG emissions by up to 2,660 million tons of CO₂e annually, which is six times the current total GHG emissions from inland freshwater aquaculture.

By assuming that aquavoltaics are preferentially deployed on aquaculture ponds with the highest electricity generation potential per unit area within each country, we estimated the proportion of

aquaculture pond area required to fully offset the carbon footprint of freshwater aquaculture across different countries (Fig.4a). Based on 2022 GHG emission levels, grid-connected aquavoltaics deployment in 63 countries could fully offset the carbon footprints of their freshwater aquaculture sectors (Fig.4b). Notably, 75% of these countries would only need to utilize less than 20% of their aquaculture pond area. As the world's largest producer of freshwater aquaculture, China produced approximately 33 million tonnes of freshwater products (including fish and bivalves) and accounted for 56% of global GHG emissions from inland freshwater aquaculture. Yet, deploying grid-connected PV systems on just 23% of its aquaculture ponds would be sufficient to offset its entire freshwater aquaculture carbon footprint through electricity generation fed into the grid (Fig.4c). In addition, the next four highest-emitting countries are India, Indonesia, Vietnam and Bangladesh, each of which would require only between 6.8% and 18.9% of their aquaculture pond area to fully offset the carbon footprint of their freshwater aquaculture sector (Fig.4d-g) in the baseline year of 2022.

Continued expansion of inland freshwater aquaculture as projected ⁸ is projected to increase GHG emissions in the sector 25% to 34% by 2050 under three socioeconomic scenarios: towards sustainability (TSS), stratified societies (SSS), and business as usual (BAUS) (Supplementary Note 1). However, even under the highest-emission BAUS scenario, only a relatively small additional proportion of aquaculture pond area would be required across countries to fully offset these future emissions. For the top five countries with the highest carbon footprint from freshwater aquaculture, the required increase in pond area ranges from 2% to 8%. If a higher PV coverage is applied, the proportion of aquaculture pond area needed would decrease even further (Supplementary Table 6). These results suggest that aquavoltaics offer a feasible pathway to offset carbon footprint from freshwater aquaculture.

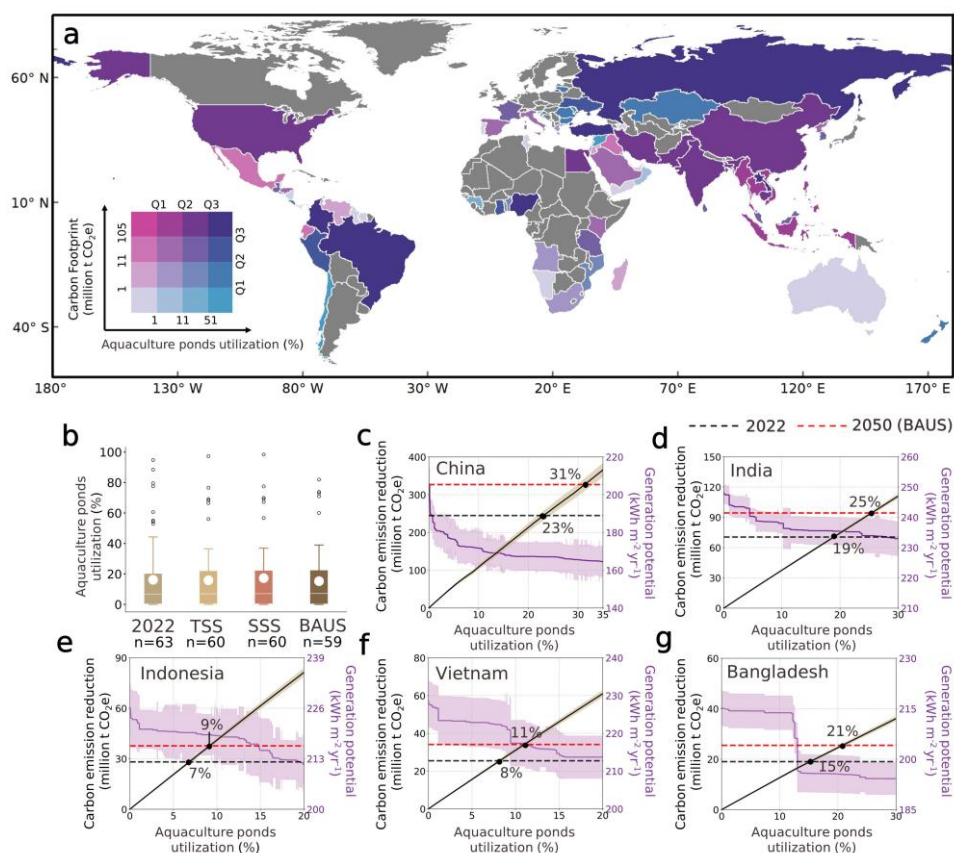


Fig. 4 Carbon footprint of national freshwater aquaculture and the required scale of aquavoltaics deployment for emission offsetting. (a) GHG emissions from freshwater aquaculture by country in 2022, and the required aquaculture pond utilization for grid-connected aquavoltaics deployment. (b) Comparison of the required aquaculture pond utilization for aquavoltaics deployment to offset emissions in 2022 and 2050 under three socioeconomic scenarios (SSS, TSS, and BAUS). The white line within the boxplots indicates the mean, while the white dot represents the median of the boxplots. n indicates the number of countries where the required utilization does not exceed 100%. (c-g) GHG emission reduction trajectories for major freshwater aquaculture-emitting countries as a function of the required aquaculture pond utilization percentage. The white line within the boxplots indicates the mean, while the white dot represents the median of the boxplots. n indicates the number of countries where the required utilization does not exceed 100%.

Discussion

Our global mapping of aquavoltaics reveals a rapidly expanding industrial paradigm integrating aquaculture and solar power. Notably, over 95% of aquavoltaics deployment has occurred in China, exhibiting an expansion rate and scale far exceeding other types of WSPV globally. This dominance is largely driven by China's 2022 policy that restricts PV construction on open water bodies such as rivers, lakes, and reservoirs due to concerns over flood control, water supply security, and ecological protection²². In contrast, numerous provincial governments have issued incentive policies and regulatory frameworks that actively promote 'fishery-solar' integration, further accelerating national aquavoltaics growth. In Southeast Asia, aquavoltaics is also developing rapidly, emerging as a key region for pilot and early commercial applications. Vietnam, for example, has launched several pilot projects that use PV-powered systems to support shrimp and pangasius aquaculture. In the Americas, early-stage demonstrations have begun with small-scale aquavoltaic pilots in Chile³⁵. In Europe, deployments have focused on marine aquaculture rather than inland aquaculture ponds, exemplified by offshore cage-based pilot projects in Norway³⁶. In most other regions, however, aquavoltaics expansion remains limited. In countries such as the United States, floating PV is more commonly deployed on water bodies that are not associated with aquaculture, leaving substantial aquavoltaics potential yet to be explored³⁷.

Despite aquavoltaics dominating the global WSPV sector, its potential is often underestimated outside of China and a few others like Laos. Our research reveals the potential for aquavoltaics to generate generation capacity equivalent to 44% and 28% of the global potential for FPV deployed on reservoirs ($9,434 \pm 29$ TWh yr⁻¹) and lakes ($14,906$ TWh yr⁻¹) respectively^{18,19}. Its advantage over FPV on these water bodies is particularly pronounced in South and Southeast Asia. In Indonesia, the Philippines, and Bangladesh, aquavoltaics has the potential to generate over ten times the output of FPV on reservoirs¹⁹. With electricity demand rising rapidly due to economic growth³⁸, these countries can use aquavoltaics as a land-efficient option for expanding low-carbon power supply while maintaining food production. In Vietnam, where electricity demand and pressure for agricultural land are increasing^{39,40}, aquavoltaics could supply nearly half of 2050 demand as it approaches 1,200 TWh⁴¹. Similar food-energy synergies are also possible in Africa through aquavoltaics. Although the continent currently produces less than 2% of global aquaculture, rising fish demand and declining wild fisheries are expected to drive substantial growth in inland pond systems, creating new surface area for PV⁴². At the same time, many African countries face chronic electricity shortages, high energy costs for cold storage and processing, and

elevated exposure to food and energy price volatility ⁴³. Aquavoltaics can, as such, enhance the sustainability and resilience of emerging aquaculture systems by improving energy access without competing for land. As pond-based aquaculture expands, increasing suitable water surface will expand Africa's long-term aquavoltaics potential, simultaneously supporting nutrition, rural livelihoods and clean energy development.

Integrating freshwater aquaculture with solar power, and exporting excess PV electricity to the grid, can further reduce the carbon footprint of freshwater aquaculture by offsetting fossil fuel generation. At the global scale, our results indicate that aquavoltaics has the potential to offset a large share of freshwater aquaculture's GHG emissions and the achievement of global emission targets ⁴⁴. However, this potential is not uniform across countries. Most aquaculture-producing countries globally could offset their freshwater aquaculture carbon footprint by deploying aquavoltaics on less than one-fifth of their existing aquaculture ponds. Conversely, our results indicate that in countries such as Nigeria and Colombia, aquavoltaics appears insufficient to fully compensate for sectoral emissions. This could stem from the underestimation of pond area in these countries by satellite mapping, which often prioritizes clustered ponds over spatially isolated ponds ²⁵. In addition, deploying aquavoltaics on smaller or dispersed ponds typically yields lower potential due to reduced grid integration benefits and higher unit electricity generation costs associated with initial investment and maintenance. Given their limited contribution to overall electricity generation potential, excluding these ponds from assessments has minimal impact on the conclusions. At the farm level, electricity from aquavoltaics can meet the needs of most farm operations—including aeration, pumping, and water treatment—thereby promoting aquaculture electrification, lowering operating costs, and improving production efficiency and product quality ⁴⁵. For example, evidence shows that PV-powered aeration, recirculating water systems, and light emitting diodes (LED) lighting can significantly boost the productivity of aquatic organisms in intensive aquaculture systems ^{46,47}.

In addition, the installation and operation of aquavoltaics can influence aquaculture production, potentially increasing operational risks for farmers. A meta-analysis of existing studies on aquavoltaic production indicates considerable variation in the impact of PV module coverage on fish growth conditions and survival rates, with unit-area yield changes ranging from -13.67% to 56.45% ¹⁶. These differences likely reflect variations in production environments, the species cultured, and feed management practices ¹⁶. Overall, the shading effect of PV panel has documented effects light intensity in ponds, potentially altering growth cycles, habitat conditions, and disease control for aquatic organisms ⁴⁸. The reduction of light availability may also suppress the growth of photosynthetic plants and algae, exerting negative impacts on herbivorous fish through trophic cascade effects ^{13,14}. Particularly under high coverage scenarios, PV panels may limit surface water mixing by restricting wind and current disturbances, thereby lowering dissolved oxygen levels and adversely affecting fish production ⁴⁹. Furthermore, PV installations may alter temperature distributions within aquaculture facilities. In summer, panels can alleviate high temperature stress on aquatic organisms by providing shade ⁵⁰. Given that optimal temperature ranges and light requirements vary across aquatic species, balancing aquaculture productivity and electricity generation requires careful system design, such as optimizing PV coverage rates. Species selection should also be tailored to local aquavoltaic conditions, especially favoring shade-tolerant aquatic products suited to regional conditions. Such integrated strategies can effectively meet market demands while significantly enhancing the economic benefits of aquaculture

production.

Moreover, implementation of aquavoltaics often reveals limitations in the technical adaptability and management capacity of conventional aquaculture systems^{48,51}. Dense PV installations may interfere with certain aquaculture activities such as harvesting, particularly in fixed aquavoltaic systems where PV modules and supporting structures are mounted directly in water. Globally, some aquavoltaic projects—including many in China—have tended to prioritize energy generation over aquaculture productivity, potentially undermining production and fisheries resource management⁵². The sustainable development of aquavoltaics therefore depends on a comprehensive set of project standards. These standards include setting minimum installation heights and maximum PV coverage to ensure that aquaculture activities remain unhindered, thus providing strong support for the sustainable development of aquavoltaics while promoting efficient resource utilization and ecological protection.

Overall, our assessment of aquavoltaics generation potential underscores its immense promise in promoting sustainable energy solutions, delivering co-benefits across the water-food-energy nexus, and accelerating the transition toward a more climate-resilient world. However, achieving the full potential of aquavoltaics requires addressing organizational, financial, and crucial grid-integration challenges. Crucially, the favorable location of many aquaculture ponds in existing electricity infrastructure regions mitigates traditional long-distance transmission constraints and supports more reliable grid connection, an advantage often not shared by FPV deployed on more remote reservoirs or lakes. Furthermore, coupling aquavoltaics with energy storage systems can effectively smooth power output and enhance grid stability. As aquaculture continues to grow and intensify over the coming decades, energy consumption and greenhouse gas emissions from this sector are likely to increase significantly. The dual capacity of aquavoltaics to produce both food and electricity offers important contributions to sustainable land use and climate change mitigation, holding particular relevance for rural development and poverty alleviation in developing countries where aquaculture is widespread.

Methods

Mapping of global aquavoltaics spatiotemporal distribution

Significant progress has been made in mapping the spatiotemporal distribution of WSPV at national and global scales^{17,23,24}. However, most existing efforts have focused on WSPV mapping without category-specific analysis, thereby overlooking the rapid and widespread expansion of aquavoltaics in recent years. To address this gap, we integrated two publicly available 10-m resolution global WSPV datasets^{23,24} with multi-source geospatial data to develop a framework for enhancing the mapping accuracy of existing products and generating the first global inventory of aquavoltaics. This framework includes three components: misclassification error correction, installation year identification, and aquavoltaics identification (Supplementary Fig. 7).

Globally, previous studies often relied on historical water surface masks with spectral indices for rapid WSPV detection. However, spectral similarities between densely arranged rooftop PV and WSPV, combined with misidentification of PV plants deployed on lands converted from former water bodies (e.g., cropland expansion, or coastal reclamation)¹⁷. To correct these errors, we used the 10-m resolution

Open Buildings 2.5D temporal dataset⁵³ to exclude WSPVs overlapping with building footprints. To further exclude agrivoltaics from the integrated WSPV dataset, we employed a rule-based classification approach based on multiple spectral indices developed by Xia et al.²⁴. We used 2023 Sentinel-2 annual mean composites with thresholds (Shortwave Infrared (SWIR)1 > 0.07, the Normalized Difference Blue/Green Blueness Index (BI_{green}) > -0.07, the Normalized Difference Blue/Red Blueness Index (BI_{red}) > -0.05), the Normalized Building Index (NDBI) > 0.02, and the Normalized Shortwave-infrared Difference Bare Soil Moisture Index (NSDSI) > 0.12) to distinguish WSPV, as agrivoltaics typically fail these criteria distinct vegetation spectral signatures. Beyond the SWIR1 band provided in the Sentinel-2 surface reflectance product, the calculation formulas for the four additional indices are computed as^{24,54}:

$$BI_{green} = \frac{\rho_{Blue} - \rho_{Green}}{\rho_{Blue} + \rho_{Green}} \quad (1)$$

$$BI_{red} = \frac{\rho_{Blue} - \rho_{Red}}{\rho_{Blue} + \rho_{Red}} \quad (2)$$

$$NDBI = \frac{\rho_{SWIR1} - \rho_{Nir}}{\rho_{SWIR1} + \rho_{Nir}} \quad (3)$$

$$NSDSI = \frac{\rho_{SWIR1} - \rho_{SWIR2}}{\rho_{SWIR1} + \rho_{SWIR2}} \quad (4)$$

where ρ_{Blue} , ρ_{Green} , ρ_{Red} , ρ_{SWIR1} , ρ_{SWIR2} are the reflectance values of the Blue, Green, Red, SWIR1, and SWIR2 bands of the Sentinel-2 imagery, respectively.

Additionally, we used global coastline dataset with 30-m resolution (GCL_FCS30)⁵⁵ and visual interpretation to remove ground-mounted PV plants on reclaimed coastal lands, especially prevalent in Japan and South Korea.

To identify installation years of each WSPV plant, we prioritized validated construction dates from 2020 China PV plant dataset produced by Chen et al.⁵⁶ for most WSPVs in China. For the rest WSPVs lacking reference dates, we used Normalized Difference Photovoltaic Index (NDPI) developed by Ren et al.²³ with long-term Landsat 5/7/8 imagery to estimate installation years, which often exhibit a marked increase following the construction of WSPV. The NDPI is computed as:

$$NDPI = \frac{\rho_{SWIR1} - \text{Max}(\rho_{Green}, \rho_{Red}, \rho_{Blue}, \rho_{Nir}, \rho_{SWIR2})}{\rho_{SWIR1}} \quad (5)$$

where ρ_{Blue} , ρ_{Green} , ρ_{Red} , ρ_{SWIR1} , ρ_{SWIR2} are the reflectance values of the Blue, Green, Red, SWIR1, and SWIR2 bands of the Landsat 5/7/8 imagery, respectively.

By integrating annual NDPI time series (2000–2023) with the LandTrendr algorithm, we detected temporal breakpoints in NDPI trajectories to infer installation years. To reduce sensitivity to outliers and noise, the NDPI transition threshold ($NDPI_{\Delta}$) was defined as the average of the 95th and 5th percentile NDPI values during the study period²³. Specifically, if a given year's NDPI exceeded $NDPI_{\Delta}$, it was designated as the installation year for that pixel. We then assigned the most frequent pixel-level year within each WSPV plant as the plant-level deployment year. For WSPV plants without detectable NDPI transitions, we manually interpreted multi-year median Landsat composites to estimate the installation year.

For aquavoltaics identification, we integrated two global reservoir and lake datasets (GRanD⁵⁷ and

GeoDAR ⁵⁸) to exclude WSPVs located on reservoirs or lakes. Additionally, GCL_FCS30 data ⁵⁵ was used to remove offshore WSPV. The remaining WSPVs were classified as aquavoltaics, except in Japan, where small, scattered ponds primarily used for agricultural irrigation rather than aquaculture ⁵⁹. Through this multi-stage process, we produced the first global aquavoltaics dataset at 10-m resolution, enabling detailed spatial and temporal analysis of aquavoltaics deployment.

Assessment of power generation and carbon mitigation potential

This study evaluated the power generation potential of over 50,000 km² of global aquaculture ponds, with their locations derived from the global LCAP dataset ²⁵. This dataset utilized edge detection and morphological analysis on the Google Earth Engine platform to create the first global spatial distribution of LCAP at 10-m resolution, based on Sentinel-2 time-series imagery from 2020.

In selecting suitable aquaculture water bodies for power generation, we followed previous biodiversity conservation criteria ¹⁸ and excluded aquaculture ponds located within high ecological value protected areas based on the World Database on Protected Areas ⁶⁰. Across more than 250,000 aquaculture pond polygons, we applied the pvlib-python library (version 0.12.0) to estimate annual electricity production of aquavoltaics ²⁶ (Supplementary Fig. 8). This open-source Python library provides a comprehensive set of functions and classes to simulate hourly theoretical outputs of solar PV systems, incorporating external conditions such as climate variables and the specifications of modules and inverters. Climate variables influencing electricity production include solar radiation, temperature, and wind speed. Solar radiation data (direct and diffuse) under all-sky conditions were obtained from the SYN1deg product ⁶¹, a three-level satellite product provided by the Clouds and the Earth's Radiant Energy System (CERES) project, offering hourly averaged shortwave downward flux on a 1° × 1° global grid. Temperature and wind speed data were sourced from the ERA5-Land hourly reanalysis product, provided by the European Centre for Medium-Range Weather Forecasts (ECMWF) on a 0.1° × 0.1° global grid ⁶². The climate data used in this study cover the period from 2001 to 2023. In addition, digital elevation dataset used for calculating the solar elevation were obtained from the 30-m Shuttle Radar Topography Mission (SRTM) dataset ⁶³.

Based on current technical level of the PV industry and predominant aquavoltaics configurations, this study applied unified assumptions for aquavoltaics systems. PV modules were chosen as SunPower SPR-315E-WHT with 19% peak efficiency, paired with TB8000SHU (240 V) inverters ^{19,64}. For temperature model parameters in the model, aquavoltaics systems adopted open-rack mounting and glass/glass module structure. Moreover, aquavoltaics systems employed widely used fixed-tilt designs, with panels oriented south in the Northern Hemisphere and north in the Southern Hemisphere (Supplementary Fig. 9). Optimal tilt angles and inter-row spacing were determined for maximum power generation. Following the third-order polynomial formulation relating optimal tilt angle to latitude, as proposed by Jacobson et al. ⁶⁵, the optimal tilt angles for the northern and southern hemispheres were calculated using equations (6) and (7), respectively.

$$\Sigma = 1.3793 + \varphi \cdot (1.2011 + \varphi \cdot (-0.014404 + \varphi \cdot 0.000080509)) \quad (6)$$

$$\Sigma = -0.41657 + \varphi \cdot (1.4216 + \varphi \cdot (0.024051 + \varphi \cdot 0.00021828)) \quad (7)$$

where Σ represents the optimal tilt angle of the PV panel; φ represents the latitude of the PV panel.

To minimize shading between adjacent panel arrays, inter-row spacing was computed using the method proposed by Masters ⁶⁶, which can be computed as:

$$D = L \cdot \cos \Sigma + L \frac{\sin \Sigma}{\cos \beta_n} \cos \varphi_s \quad (8)$$

where D represents the inter-row spacing; L represents the width of PV panel; Σ represents the optimal tilt angle of the PV panel; β_n and φ_s represent the solar altitude and solar azimuth at local standard time 3:00 p.m. in the winter solstice of 2023, respectively.

To meet daily fishery operational requirements, we set the height of the PV module center above ground at 2.5 m, following the Compiling requirements for planning of photovoltaic agricultural park (Jiangsu Provincial Standard) ⁶⁷. This standard, established by Jiangsu—China's leading province in aquavoltaics by area—requires a minimum vertical clearance of 2 meters between the lowest edge of a PV module and the average annual aquaculture water level. Designed for operational efficiency, this regulation from Jiangsu offers a practical model for global aquavoltaic applications ⁶⁷. Additionally, we accounted for water albedo effects on PV generation ⁶⁸, incorporating water albedo as a model input parameter. Albedo for aquaculture pond surfaces was calculated using the model proposed by Dvoracek & Hannabas ⁶⁹, which can be computed as:

$$\rho = c^{r \cdot \sin(\alpha) + 1} \quad (9)$$

where ρ represents the albedo for aquaculture pond surfaces; α represents the solar elevation; c represents the water color coefficient; r represents the water wave roughness coefficient; Aquaculture pond surfaces were classified as muddy water with no waves in this study, with a water color coefficient of 0.19 and a surface roughness coefficient of 0.29 ⁶⁹.

To determine the coverage of PV panels in aquaculture ponds, defined as the ratio of the panels' vertical projection area to the pond's water surface area, we adopted the threshold of 40%. However, due to the wider optimal row spacing required for efficient power generation in high-latitude regions ⁷⁰, the PV coverage typically falls below the 40% regulatory threshold. Accordingly, we adopted the coverage determined by the optimal row spacing in these areas for maximizing energy output. To assess the contribution of aquavoltaics to urban-scale electricity demand, we aggregated global city-level electricity demand using the 2019 revised electricity consumption dataset ³¹. This dataset used a top-down approach to estimate global electricity consumption on a 1 km × 1 km grid based on calibrated nighttime light data.

In this study, we adopted the combined margin (CM) baseline approach proposed by the United Nations Framework Convention on Climate Change (UNFCCC) to quantify the carbon mitigation potential of deploying aquavoltaics across different countries. Carbon emission reductions were calculated by multiplying the electricity generated by aquavoltaics with country-specific CM grid emission factors ⁷¹. The CM factor was calculated as a weighted average of the operating margin (OM) and build margin (BM), where the OM represents the emission intensity of existing power plants that are most likely to be affected by reduced generation (typically those with higher operational costs or carbon intensity), while the BM represents the emission intensity of newly built power plants that might be avoided due to increased renewable deployment ⁷². This approach captures both short-term and long-term marginal impacts of renewable electricity on national grids, providing a more comprehensive measure of the carbon mitigation potential of aquavoltaics. In our analysis, we utilized national-level CM grid emission factors published by the UNFCCC in 2022 ⁷³.

Analysis of the carbon footprint of global freshwater aquaculture

The GHG emission intensity of major global freshwater aquaculture groups (excluding aquatic plants, i.e., fish-general, bivalves, catfish, cyprinids, shrimps, tilapia, etc.) was derived from Shen et al.⁸. Their analysis employed a life-cycle assessment approach with a “cradle-to-farm-gate” system boundary, encompassing GHG emissions from feed production (crop cultivation, energy and fertilizer manufacturing, and non-crop feed processing), energy consumption, and the aquatic environment. Building on prior studies^{5,74}, Shen et al. compiled a global inventory of freshwater aquaculture GHG emissions, integrating measurements from 107 sites and updated datasets to enhance representation of underrepresented species to improve the accuracy of emission estimates⁸. Country-level freshwater aquaculture production data for 2022 were obtained from the FAO FishStatJ database⁷⁵. For each country, the carbon footprint (CF) was calculated as follows:

$$CF = CF_{feed} + CF_{energy} + CF_{aquatic\ environment} = \sum_{i=1}^n EI_i \cdot P_i \quad (10)$$

where EI_i is the emissions intensity (kg CO₂e per kg) of a specific species group, and P_i is the production volume (in tonnes) of that species group in the country.

Assuming global fish protein demand is met solely by freshwater aquaculture, Shen et al. projected global GHG emissions from freshwater aquaculture in 2050 under three socioeconomic scenarios: towards sustainability, business as usual, and stratified societies⁸ (Supplementary Note 1). We assumed that each country’s share of global GHG emissions from freshwater aquaculture remains constant over time and allocated the projected 2050 global GHG emissions to individual countries for each scenario, producing country-level estimates for each scenario.

Data availability

The 2023 global WSPV dataset was obtained from <https://eetingtinghe2011.projects.earthengine.app/view/globalwaterpv2023>. The 2020 global WSPV dataset was obtained from <https://zenodo.org/records/8211179>. Landsat-5/7/8 and Sentinel-2 surface reflectance data, the Open Buildings 2.5D temporal dataset, and the SRTM digital elevation dataset are available via GEE platform (<https://code.earthengine.google.com>). The GCL_FCS30 dataset is openly available via Zenodo at <https://doi.org/10.5281/zenodo.13943679>. Solar radiation data are available from the CERES SYN1deg Level-3 product at <https://ceres.larc.nasa.gov/data/#syn1deg-level-3>. Temperature and wind speed data are provided by the ERA5-Land dataset at <https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-land?tab=overview>. The GGrAnD data was obtained from <https://sedac.ciesin.columbia.edu/data/collection/grand-v1>, and the GeoDAR data is available at <https://doi.org/10.5281/zenodo.6163413>. The global LCAP dataset is available at <https://www.scidb.cn/en/detail?dataSetId=5214b31810db49b8bbfdc1243ce5c56b>. The national-scale electricity demand dataset for 2019 is available at <https://database.earth/energy/electricity-demand/2019>. The global gridded revised real gross domestic product and electricity consumption dataset for 2019 are available at <https://doi.org/10.6084/m9.figshare.19517272.v1>. National-level CM grid emission factors are provided by the UNFCCC at <https://unfccc.int/documents/461676>.

Code availability

All code used in this study is publicly available at <https://zenodo.org/records/17863816>.

Acknowledgments

This research was supported by the National Natural Science Foundation of China under Grant

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